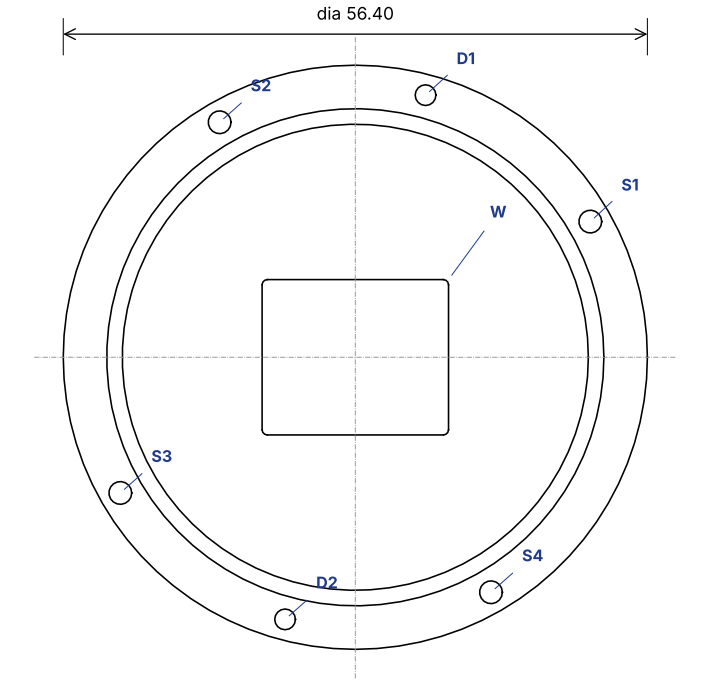
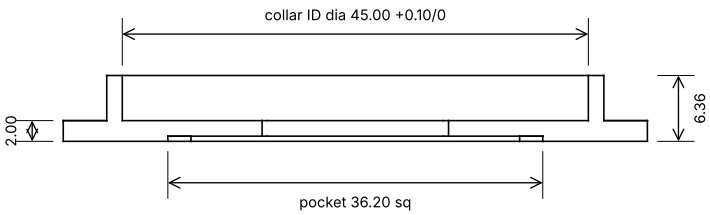
**TOP VIEW (looking down the baffle axis)**




**SECTION A-A (through the axis, seen from -Y)**

## FEATURE TABLE

**O** Flange dia 56.40, 2.00 thick; lower face is the mounting face, flat within 0.02.

**C** Collar: ID dia 45.00 +0.10/0, OD dia 48.00, top 6.36 above the mounting face; concentric with the window centre within 0.05.

**P** Pocket in the mounting face for the slit plate: 36.20 × 36.20, with 4x corner reliefs dia 3.175 (1/8 in tool), centres at X +-17.10 Y +-17.10, same depth as the pocket, so a square-cornered plate seats. Depth equal to the measured thickness of the slit plate (0.50 nominal), so the plate's lower face is flush with the mounting face within 0.01. Carthage bonds the plate into the pocket.

**W** Window over the film field: 18.00 × 15.00 thru the flange, corners R0.5 to R1.6 (shop's choice), long side along X. Assumes films 3.0 x 14.0 per slit (1.5 beyond each side of a slit, 2.0 beyond each end); revise when Luxel gives the film margin.

**S1** dia 2.20 thru (M2 screw). X 22.69 Y 13.10

**S2** dia 2.20 thru (M2 screw). X -13.10 Y 22.69

**S3** dia 2.20 thru (M2 screw). X -22.69 Y -13.10

**S4** dia 2.20 thru (M2 screw). X 13.10 Y -22.69

**D1** dia 2.00 +0.012/0 thru (dowel, slip fit). X 6.78 Y 25.31

**D2** dia 2.00 +0.012/0 thru (dowel, slip fit). X -6.78 Y -25.31

**F** All surfaces visible from the window and the collar bore: matte, then black for flight (as the H02 cone). As machined is acceptable for this unit.

## NOTES

Units: millimetres. Third-angle projection. Tolerances unless stated: features and hole positions +-0.1; outline +-0.2.
Threads: ISO metric coarse, 6H. Holes marked INSERT are tapped for stainless free-running helical coil inserts, length 1.5D, supplied and installed by the shop; drill and tap to the insert maker's data. Inserts flush to 0.25 below.
Every drilled and tapped hole runs through to an open volume (venting). Break sharp edges 0.1-0.3; deburr all holes.
Finish: as machined for this unit. Please also quote: clear anodize MIL-A-8625 Type II Class 1, with the H01 host face and the panel/angle faces under rivets masked and conversion coated MIL-DTL-5541 Type II Class 3 (electrical bonding).
Assembly is by Carthage: fasteners locked and torqued per the SLANT assembly procedure (to be issued).
Fit-check unit built to the flight layout. Material certs appreciated; no inspection report required.
Model: the STEP file of the same part number governs any dimension not shown here.
Mounts to the optical bulkhead with 4 M2 screws into self-clinching nuts and 2 dowels; removable from above.

| | |
|---|---|
| PART | **SLANT-H09 Filter cartridge** |
| MATERIAL | 6061-T6 plate, 3/8 in stock |
| QTY | 1 |
| MASS (model) | 13.1 g |
| PROJECT | SLANT 1U housing, assembly rev 16 |
| OWNER | Carthage College, Space Sciences |
| DRAWN / CHECKED | from model 2026-10-08 / _____ |
| SHEET REV | D |